NEXON — Agentic AI Operating System

Information system Project Report | Ritik Raj | 25IM60R12

OVERVIEW

NEXON is a full-stack, cross-platform desktop AI operating system built from scratch. It runs entirely on a local machine — no mandatory cloud services — using Ollama's locally hosted large language models as the primary intelligence engine. The application is packaged as an Electron desktop app (Windows and macOS), with a Python FastAPI backend and a modern HTML/CSS/JavaScript frontend that delivers a futuristic, animated interface. The core idea is to combine voice, facial emotion, hand gestures, and natural language into a single unified system that understands not just what the user says, but how they are feeling when they say it, and responds accordingly.

TECHNICAL ARCHITECTURE

Backend: FastAPI serves 25+ REST endpoints and two WebSockets. SQLite (via SQLAlchemy) stores sessions, messages, user preferences, clipboard history, memory nodes, usage patterns, personality profiles, and gesture macros. All data persists between restarts. A database migration script handles schema upgrades without data loss.

LLM Engine: A unified *NexonLLM* class wraps Ollama (primary, local) with automatic fallback to Groq cloud API. The system prompt is engineered to suppress raw JSON from appearing in chat responses, and a fast rule-based classifier prevents parallel agents from triggering on simple greetings.

Frontend: Three resizable columns separated by draggable dividers. The left column holds a live MediaPipe camera feed with face mesh overlays, an emotion mirror avatar, emotion and gesture cards, voice stress bar, and predictive suggestions. The center column contains a Three.js 3D neural sphere of 250 particles that reacts to speech state, a mic button, and a 48-bar audio waveform visualizer. The right column is the chat interface with markdown rendering, emotion-tagged message bubbles, and agent action badges.

KEY FEATURES

- Multimodal Context Injection: every prompt sent to the LLM is enriched with the user's current facial emotion (with confidence %), detected hand gesture, and voice stress score. The user sees only their original message in chat.
- 10 Specialized Agents: Email (Gmail SMTP + LLM-drafted bodies), Calendar, File creation/conversion, Screen control & OCR, Web scraping & search, Data analysis, Messaging (SMS/WhatsApp/Slack/Discord), Smart home, Finance tracking, and Productivity.
- Parallel Execution: compound commands like "send email AND book meeting AND screenshot" run all agents simultaneously via asyncio.gather() with live WebSocket status streaming.
- Long-Term Memory Graph: sentence-transformers embeddings stored in SQLite. Every conversation auto-extracts facts, preferences, and decisions. Relevant memories are injected as context into future prompts.
- Predictive Intent Engine: mines time-of-day, day-of-week, emotion, and action-sequence patterns to suggest next actions before the user asks.
- Personality Style Engine: learns formality, directness, vocabulary level, and emoji preference from every message, shaping LLM response tone over time.
- Human Female Voice TTS: selects Samantha (macOS) or Google UK English Female in priority order. Speech rate and pitch adjust per emotion. A background SpeechRecognition listener detects "Hey NEXON" hands-free.
- Biometric Auth: on-device face recognition (face_recognition library) with a splash screen on startup. Recognized users get their own memory graph and personality profile automatically loaded.

- Voice Stress Analysis: librosa extracts pitch variance, RMS energy, speech rate, and spectral centroid from audio to classify the speaker as calm, stressed, excited, or confident.
- Gaze Tracking: MediaPipe iris landmarks (468–477) drive a screen-space gaze cursor with 1.5-second dwell activation on buttons and edge-scroll in the chat area.

CHALLENGES SOLVED

The most persistent technical problem was audio transcription. Electron's MediaRecorder produces WebM blobs that ffmpeg could not parse, causing 500 errors on every voice attempt. The fix was to bypass MediaRecorder entirely: the recorder now uses the Web Audio API ScriptProcessor to capture raw Float32 PCM samples at 16 kHz and encodes them into a proper WAV file with a 44-byte RIFF header before sending to Whisper. This eliminated the ffmpeg dependency and reduced transcription time from 15 seconds (base model, WebM) to 1–2 seconds (tiny model, WAV). A second major issue was emotion detection bias: the heuristic was classifying neutral faces as SAD because the smile-score threshold was too sensitive. Recalibrating around face-height-normalized measurements and requiring much stronger signals — with a 14-frame smoothing buffer — fixed the accuracy. Biometric face recognition also failed initially because the splash screen was capturing a CSS-mirrored canvas frame; removing the mirror transform from the capture path resolved it.

OUTCOME

NEXON delivers a working AI operating system that runs entirely on a consumer laptop. On an Apple M4 MacBook Air it achieves 1–2 second voice transcription, 3–8 second LLM responses, 100 ms face detection, and real-time gesture classification at 7 fps. All core features — chat, voice, emotion detection, gesture recognition, long-term memory, personality learning, parallel agents, predictive suggestions, and biometric authentication — function without any internet connection or paid API subscription. Optional integrations for email (Gmail), SMS (Twilio), and faster inference (Groq) activate by adding credentials to a single *.env* file. The project comprises 25 Python modules, 10 JavaScript renderer files, 2 Electron process files, and 25+ API endpoints across roughly 8,000 lines of original code.